DIGITAL LITERACY IN RESEARCH

Navigating Research in the Age of Digital Tools and Generative AI

Learning Objectives

AT THE END OF THIS SESSION

- Menjelaskan peran literasi digital dalam proses penelitian modern.
- Memetakan dukungan digital tools pada setiap tahap research lifecycle.
- Membedakan digital skill dengan digital research literacy.
- Memahami research gap dan beberapa bentuknya.
- Menggunakan GenAI untuk eksplorasi gap secara terarah.
- Memverifikasi output GenAI secara kritis sebelum digunakan sebagai dasar penelitian.

Research Has Changed

Prinsip ilmiah tetap, tetapi lingkungan penelitian berubah.

Conventional research

Physical library
Printed journals
Manual data collection
Small / structured datasets
Standalone software
Mostly individual workflow

Contemporary research

Digital scholarly databases
Open data & repositories
APIs, web and platform data
Large / unstructured datasets
Cloud collaboration
AI-assisted research

Research Topic

↓

Literature

↓

Research Gap

↓

Data & Analysis

↓

Publication

Digital tools + critical judgment

Research competency now includes the ability to work critically inside a digital research environment.

What Is Digital Literacy?

Digital literacy is not merely the ability to use technology.



Digital literacies: the set of technical, visual, behavioural, critical and social skills required to use current technologies that enable users to learn, communicate, socialise and contribute in the digital space

Dalam konteks akademik: menemukan, menilai, menggunakan, menghasilkan, dan mengomunikasikan informasi digital secara kritis, efektif, dan etis.

Reference: Reyna, Hanham & Meier (2018). A framework for digital media literacies for teaching and learning in higher education.

Digital Skill ≠ Digital Literacy

Digital Skill

Knowing HOW

- Operate a tool
- Search information
- Use AI
- Create a chart
- Download a dataset

Digital Literacy

Knowing HOW, WHY, WHEN—and WHAT CAN GO WRONG

- Evaluate credibility
- Verify AI output
- Interpret visualization
- Assess provenance & bias
- Choose tools fit for purpose

Competent tool user ≠ competent researcher

Reference: ACRL (2016). Framework for Information Literacy for Higher Education.

Components of Digital Research Literacy

Information literacy

Find & evaluate information

Data literacy

Understand & evaluate data

Tool literacy

Select & use tools

AI literacy

Use & critique AI

Ethics literacy

Privacy, copyright, integrity

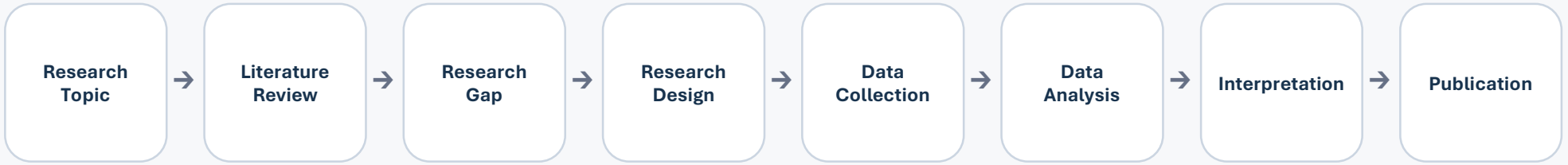
Critical literacy

Bias, limits, platform effects

RESEARCH LITERACY

Reference: Rogers (2024). Doing Digital Methods. SAGE.

The Research Lifecycle



Digital technologies can support almost every stage—but support is not the same as methodological validity.

Reference: Creswell & Creswell (2022). Research Design, 6th ed.

Digital Research Ecosystem

Problem formulation	Search, trend analysis, GenAI
Literature discovery	Google Scholar, Scopus, Web of Science
Reference management	Zotero, Mendeley
Literature mapping	VOSviewer, ResearchRabbit
Data collection	Survey tools, APIs, web scraping
Analysis	R, Python, SPSS, NVivo
Communication	Visualization, collaborative writing
Publication	Repositories, journal platforms, ORCID

Tools will change. Research principles should not.

Reference: Dawson (2020). A–Z of Digital Research Methods. Routledge.

The Tool Should Follow the Research Question

Technology-driven research

“Saya ingin menggunakan Machine Learning. Kira-kira penelitian apa yang bisa dilakukan?”

Risiko: novelty semu, method mismatch, dan pertanyaan penelitian dipaksa mengikuti tool.

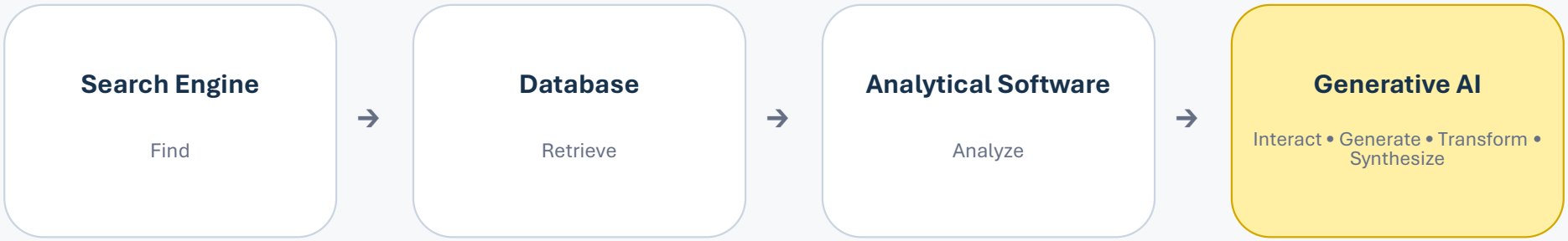
Research-driven technology

“Saya memiliki research question X. Data apa yang dibutuhkan? Metode apa yang tepat? Tool apa yang membantu menjalankannya?”

Research Question → Methodology → Method → Data → Tool

Reference: Gray & Jensen (2026). Doing Research in the Real World, 6th ed. SAGE.

From Digital Tools to Generative AI



LLMs can generate, summarize, classify, extract, transform, explain and compare—but fluency is not evidence.

Reference: Park & Choi (2024). How to harness the power of GPT for scientific research.

Technology Behind Text Generative AI

Table 1. Techniques used in generative AI

Machine Learning (ML)		A type of AI that uses data to automatically improve its performance.
Artificial Neural Network (ANN)		A type of ML that is inspired by the structure and functioning of the human brain (e.g. the synaptic connections between neurons).
Text generative AI	**General-purpose Transformers**	A type of ANN that is capable of focusing on different parts of data to determine how they relate to each other
	Large Language Models (LLM)	A type of General-purpose Transformer that is trained on vast amounts of text data.
	Generative Pre-trained Transformer (GPT)[4]	A type of LLM that is pre-trained on even larger amounts of data, which allows the model to capture the nuances of language and generate coherent context-aware text.
Image generative AI	**Generative Adversarial Networks (GANs)**	Types of Neural Network used for image generation.
	Variational Autoencoders (VAEs)	

Table 2. OpenAI GPTs

MODEL	LAUNCHED	AMOUNT OF TRAINING DATA	NUMBER OF PARAMETERS	CHARACTERISTICS
GPT-1	2018	40 GB	117 million	Capable of natural-language-processing tasks such as completing texts and answering questions.
GPT-2	2019	40 GB	1,500 million	Capable of more complex natural language processing tasks such as machine translation and summarizing.
GPT-3	2020	17,000 GB	175,000 million	Capable of advanced natural language processing tasks such as writing coherent paragraphs and generating entire articles. Also capable of adapting to new tasks with just a few examples.
GPT-4[6]	2023	1,000,000 GB (reported but not confirmed)	170,000,000 million (reported but not confirmed)	Enhanced reliability and is capable of processing more complex instructions.

Response to a prompt involves the following steps

1. Prompt is broken down into smaller units (called **tokens**) that are input into the GPT.
2. The GPT uses **statistical patterns** to predict likely words or phrases that might form a coherent response to the prompt
    1. The GPT identifies patterns of words and phrases that commonly co-occur in its prebuilt large data model
    2. Using these patterns, the GPT estimates the **probability** of specific words or phrases appearing in a given context.
    3. Beginning with a random prediction, the GPT uses these estimated probabilities to predict the next likely word or phrase in its response.
3. The predicted words or phrases are converted into **readable text**.
4. The readable text is filtered through what are known as '**guardrails**' to remove any offensive content.
5. Steps 2 to 4 are repeated until a response is finished. The response is considered finished when it reaches a maximum token limit or meets predefined stopping criteria.
6. The response is post-processed to improve readability by applying formatting, punctuation, and other enhancements (such as beginning the response with words that a human might use, such as 'Sure', 'Certainly', or 'I'm sorry').

Reference: UNESCO (2023). Guidance for generative AI in education and research

Where Can GenAI Assist Research?

Project Planning

Steps
gantt chart/timeline

Problem

Brainstorming
concept exploration

Literature

Keywords
query formulation
summarization

Design

Method discussion
critique

Data

Extraction
classification

Analysis

Coding help
explanation

Writing

Structure/outline
language editing

AI assistance ≠ AI authority

Reference: Nature (2024). How ChatGPT has changed scientists' work.

Project Planning

User

I am preparing for an educational research project, but I'm not sure how to begin tackling my topic. Could you help me consider some ways of planning out the task? My research question is 'How are GenAI tools affecting doctoral researchers in educational studies?'

Example 9-Month Gantt Chart

Task	M1	M2	M3	M4	M5	M6	M7	M8	M9
1. Define Research Scope & Refine Research Questions	■■■	■■■							
2. Literature Review & Theoretical Framework	■■■	■■■	■■■						
3. Ethical Approval (IRB) Process		■■■	■■■						
4. Research Instrument Design & Pilot Testing		■■■	■■■	■■■					
5. Data Collection (Surveys, Interviews, Focus Groups)				■■■	■■■	■■■			
6. Data Cleaning & Analysis					■■■	■■■	■■■		
7. Interpretation & Write-Up of Findings						■■■	■■■	■■■	
8. Revisions & Finalizing Thesis/Paper							■■■	■■■	■■■

Figure 1 An example 9-month Gantt chart produced by OpenAI o1-Mini

Reference: Roe, Jasper. (2025) How to Use Generative AI in Educational Research.

OpenAI o1-Mini

Certainly! Planning a comprehensive educational research project involves several key steps, from refining your research question to executing your methodology and analyzing your findings. Below is a structured approach tailored to your research question:
How are GenAI tools affecting doctoral researchers in educational studies?'

1. Define the Scope and Objectives
a. Clarify Your Research Question:
• Ensure that your research question is specific, measurable, achievable, relevant, and time-bound (SMART).
• Consider sub-questions, such as:
◦ What types of GenAI tools are most commonly used by doctoral researchers in educational studies?
◦ How do these tools impact the research process (e.g., data analysis, literature review, writing)?
◦ What are the perceived benefits and challenges of using GenAI tools?
...

Use AI to help plan and manage the project

Four Roles of AI in Scientific Research

AI as Oracle

Answers questions

AI as Surrogate

Substitutes for humans / observations

AI as Analyst

Finds patterns in data

AI as Arbiter

Ranks / evaluates / selects

More AI capability can increase productivity—and also create an “illusion of understanding.”

Reference: Messeri & Crockett (2024). Artificial intelligence and illusions of understanding in scientific research. Nature.

What Is a Research Gap?

A research gap is an area where existing knowledge is incomplete, inconsistent, insufficient, or requires further investigation.

- **Incomplete evidence**
- **Conflicting findings**
- **Insufficient theory**
- **Method limitations**
- **Under-researched context**
- **Under-represented population**

Reference: Robinson et al. (2013). A framework for determining research gaps during systematic reviews.

“No One Has Studied This” Is Not Enough

Weak gap statement

“There are limited studies about AI adoption in SMEs in Indonesia.”

Question: So what? Why does the absence matter theoretically or practically?

Stronger gap logic

What we know
+ What we do not know
+ Why it matters
= A defensible research gap

A gap needs evidence and justification—not just scarcity.

Reference: Research gap statement and justification in higher education research (2023).

Types of Research Gaps

Evidence gap

Findings are inconsistent or insufficient

Methodological gap

Dominant methods have limitations

Knowledge gap

An aspect remains poorly understood

Context / population gap

Specific contexts or groups are under-studied

Empirical gap

A claim/theory is not sufficiently tested

Implementation gap

Knowledge exists but application remains unclear

Theoretical gap

Current theory cannot explain the phenomenon

Reference: Snyder (2019). Literature review as a research methodology. Journal of Business Research.

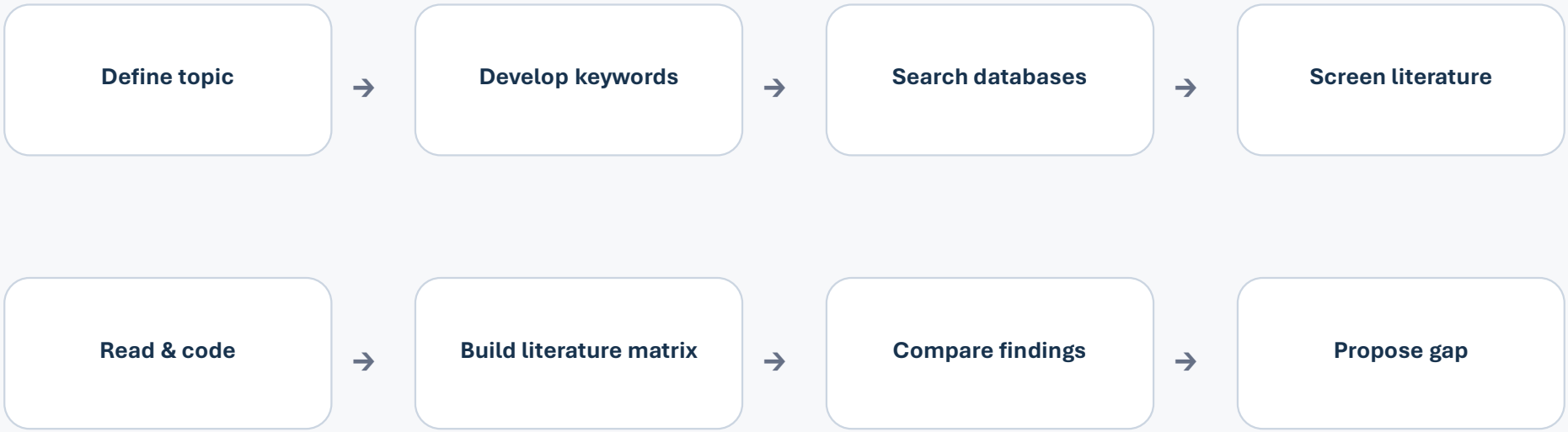
Where Do Research Gaps Come From?

- Contradictory findings across studies.
- Limitations explicitly acknowledged by prior research.
- Future research recommendations—but these must still be judged critically.
- Theory that fails to explain a phenomenon or boundary condition.
- Research concentrated in certain countries, sectors, or populations.
- Dominance of one method, design, dataset, or timescale.
- New phenomena caused by technological, social, regulatory, or behavioral change.

A gap is discovered through synthesis—not by reading one paper.

Reference: Imperial College London Library. Finding research gaps in the literature.

Traditional Gap Identification



Challenges: literature volume • time • cognitive load • synthesis complexity

Reference: Snyder (2019). Literature review as a research methodology.

AI-Assisted Gap Identification



AI can accelerate comparison. It cannot independently prove novelty.

Reference: Literature review of AI research using machine learning and ChatGPT (2024). Data and Information Management.

Good vs Poor Use of GenAI for Gap Analysis

Better use

- Generate keywords
- Compare supplied articles
- Extract limitations
- Identify contradictions
- Build literature matrices
- Generate candidate gaps

Poor use

- “Give me a novel gap” with no evidence
- Trust the model’s memory of literature
- Accept invented references
- Assume “not studied” is true
- Skip database searching
- Let AI decide novelty

Example:

Prompts for Brainstorming
- Generate three minor modifications of the research problem [insert research problem], making it more [narrow | general | original | difficult | significant | . . .].
- My research problem is [insert research problem]. Suggest some root causes of this problem.
- Generate a story based on the situation that [insert research problem], addressing [economic | health | legal | . . .] aspects

Use AI to interrogate the literature—not to replace the literature.

Reference: Paul Johannesson: Writing Your Thesis with ChatGPT, Research, Scholarship and Academic Writing in the Age of Generative AI, January 2024.

Prompting for Research Gap Analysis

C — Context

Field / topic

T — Task

What should be analyzed?

E — Evidence

What papers / data may be used?

C — Constraints

What must the model NOT do?

O — Output

How should results be structured?

“Using only the abstracts provided below, compare the studies in terms of theory, methodology, population, context and findings. Identify recurring limitations and contradictory findings. Propose candidate research gaps, but do not claim novelty unless supported by the supplied studies.”

Ground the AI in evidence you provide.

Reference: Park & Choi (2024). How to harness the power of GPT for scientific research.

Example: From Literature to Candidate Gap

Topic: AI adoption and firm performance

Study	Country	Sample	Method	
A	USA	Large firms	Survey	
B	Germany	Large firms	Survey	
C	China	Manufacturing	SEM	
D	UK	Large firms	Survey	

AI candidate gap

“Few studies examine SMEs in emerging economies.”

Is this already a research gap?

Not yet. Verify coverage, relevance, theoretical reason, and whether the gap is worth studying.

Reference: UChicago Library. Research gaps: how to identify and evaluate them.

The Hallucination Problem

Smith, J. & Lee, K. (2021). "Artificial Intelligence and Supply Chain Resilience." Journal of Strategic Logistics, 14(2), 45–62.

Looks academic. But what if the paper does not exist?

Plausibility ≠ truth

Reference: Walters & Wilder (2023). Fabrication and errors in bibliographic citations generated by ChatGPT. Scientific Reports.

AI Can Create an Illusion of Understanding

Before AI

“I haven’t read enough to know.”

Uncertainty is visible.

With fluent AI output

“The explanation sounds coherent, so I probably understand it.”

Fluency can create overconfidence.

More output does not necessarily mean more knowledge.

Reference: Messeri & Crockett (2024). Artificial intelligence and illusions of understanding in scientific research. Nature.

Verification: The Researcher Remains Accountable

1. SOURCE

Does the source exist?

2. CONTENT

Does it actually say this?

3. QUALITY

Is the source credible?

4. COVERAGE

Are important studies missing?

5. INTERPRETATION

Is the finding interpreted correctly?

6. NOVELTY

Has the alleged gap really been addressed?

AI proposes → Researcher verifies → Evidence decides

Reference: Farquhar et al. (2024). Detecting hallucinations in large language models. Nature.

Class Exercise & Key Takeaways

10–15 minute exercise

Topic: “Generative AI adoption among university students”

1. Ask AI for 5 research gaps + references.
2. Select 2 candidate gaps.
3. Verify references and claims.
4. Classify the type of gap.
5. Rewrite as: What we know → What we don’t know → Why it matters.

Takeaways

1. Digital literacy > tool proficiency.
2. Methodology should determine tools.
3. GenAI can accelerate exploration.
4. AI cannot establish novelty by itself.
5. AI proposes. Researchers verify. Evidence decides.